SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS MASTER IN CONTROL

1. Exercise

a) The function $f(x)$ that defines the system's equations is polynomial in both variables. Hence it is locally Lipschitz and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition).

b) The nullclines of the vector-field can be computed as shown below:

$$\mathscr{N}_1 = \{x : x_2 = 0\}.$$

This is the equation of a horizontal line through the origin. Moreover, all solutions above the line move towards higher values of x_1 (positive $\dot{x}_1$), and those below move towards lower values. For the second nullcline we see:

$$\mathscr{N}_2 = \{x : -x_1^3 + x_1 - x_2(1+x_1)^2 = 0\} =$$

$$= \{x : x_1 = -1\} \cup \left\{x : x_2 = \frac{x_1(1-x_1)}{1+x_1}\right\}.$$

The second nullcline is therefore composed of two branches as sketched in Figure 1.1.

Several students missed the decomposition of the nullcline in two separate branches. Others did not notice the possibility of simplifying the equation of the nullcline

c) Since $\dot{x}_2$ is a monotone decreasing function of x_2, above the second nullclines solutions move towards lower values of x_2 (South) and below the nullcline move towards higher values (North). Overall, there are 8 regions, as shown in Figure 1.1.

Several students struggled to produce a consistent picture having identified wrong nullclines in the first place

d) It is possible to see that none of the regions are foward invariant. Indeed, all regions have a part of the boundary with arrows pointing outside the region. See Figure 1.1.

e) The equilibria are at the intersection of the nullclines. In particular, there are 3 equilibria:

$$x_{e1} = [-1,0]', \quad x_{e2} = [0,0]', \quad x_{e3} = [1,0]'.$$

Several students did not compute all equilibria correctly

f) Computing the Jacobian of $f(x)$ we see:

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -3x_1^2 + 1 - 2x_2(1+x_1) & -(1+x_1)^2 \end{bmatrix}.$$

Evaluating the above expression in x_{e1} yields:

$$\left.\frac{\partial f}{\partial x}\right|_{x=x_{e1}} = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}.$$

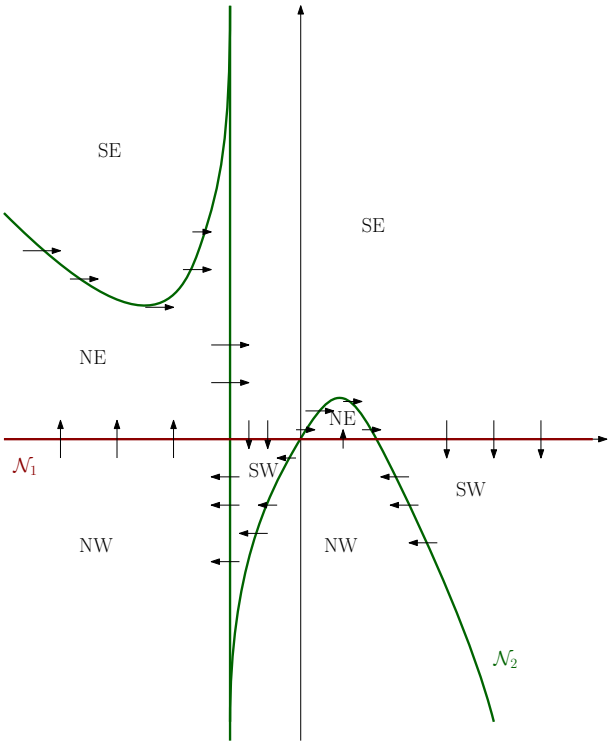


Figure 1.1 Nullclines and vector-field orientations

Eigenvalues of the above matrix are purely imaginary $\pm\sqrt{2}j$. Hence x_{e1} is not hyperbolic. Its local phase-portrait cannot be inferred from Hartman-Grossmann Theorem. Evaluating the above expression in x_{e2} yields:

$$\left.\frac{\partial f}{\partial x}\right|_{x=x_{e2}} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}.$$

The characteristic polynomial is: s^2+s-1 and the corresponding roots are real with opposite signs:

$$\lambda_{1,2} = \frac{-1\pm\sqrt{5}}{2}.$$

Hence, the equilibrium x_2 is hyperbolic and its local phase-portrait is that of a saddle point. Evaluating the above expression in x_{e3} yields:

$$\left.\frac{\partial f}{\partial x}\right|_{x=x_{e3}} = \begin{bmatrix} 0 & 1 \\ -2 & -4 \end{bmatrix}.$$

The characteristic polynomial is s^2+4s+2, which admits two real roots of negative sign in $-2\pm\sqrt{2}$. Hence this equilibrium is hyperbolic and the local phase-portrait is that of a stable node.

g) We need to analyze local stability of x_{e1} which is non-hyperbolic. To this end, let us take the mechanical energy of the system as a candidate Lyapunov function:

$$V(x) = x_2^2/2 + x_1^4/4 - x_1^2/2 + 1/4$$

Notice that $V(x_{e1}) = 0$, moreover V has a local minimum in $[-1,0]'$, hence V is positive definite in a neighborhood of x_{e1}. Taking derivatives along solutions yields:

$$\frac{\partial V}{\partial x}(x)f(x) = (x_1^3 - x_1)x_2 + x_2(-x_1^3 + x_1 - x_2(1+x_1)^2) = -x_2^2(1+x_1)^2 \leq 0$$

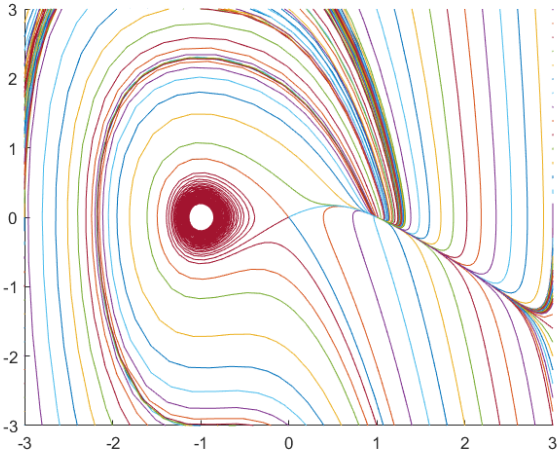


Figure 1.2 Phase portrait

Hence, x_{e1} is locally Lyapunov stable. Moreover

$$\mathrm{Ker}[\dot{V}(x)] = \{x : 1 + x_1 = 0\} \cup \{x : x_2 = 0\}.$$

If an invariant set is contained in the Kernel then any of its point need to fulfill either

$$1 + x_1 = 0 \text{ and } \dot{x}_1 = 0 \Rightarrow x_1 = -1 \text{ and } x_2 = 0$$

or

$$x_2 = 0 \text{ and } \dot{x}_2 = 0 \Rightarrow x \in \mathscr{N}_1 \cap \mathscr{N}_2$$

Hence, in a sufficiently small neighborhood of x_{e1} the largest invariant set contained in $\mathrm{Ker}[\dot{V}(x)]$ is $\{x_{e1}\}$. This proves the local asymptotic stability of x_{e1} (indeed the local phase portrait is a stable focus).

h) The global phase portrait is sketched in Fig. 1.2.
A typical error is not to pay attention to how local phase-portraits merge with each other in a way that is plausible. Computing eigenvectors for saddle equilibria is normally quite useful in this respect.

2. Exercise

a) We choose $V(x) = x_1^2 + x_2^2$ as a candidate Lyapunov function. Notice that $V(x)$ is of class $\mathscr{C}^1$, positive definite and radially unbounded. So it can work for a Lyapunov analysis both in a global or local context. Taking derivatives along solutions yields:

$$\dot{V}(x) = \frac{\partial V}{\partial x}(x) \cdot f(x) = [2x_1, 2x_2] \begin{bmatrix} x_2 - x_1 x_2^2 \\ -x_1 - x_2 x_1^2 \end{bmatrix} = -4x_1^2 x_2^2.$$

Hence, $\dot{V}(x) \leq 0$ for all $x \in \mathbb{R}^2$ and therefore $\dot{V}$ is negative semi-definite. This implies Lyapunov stability of the equilibrium $x_e = 0$.
Typical error claiming that $\dot{V}$ is negative definite, just because it is ≤ 0 and contains both state variables.

b) Notice that $\dot{V}(x)$ is not negative definite. Indeed, $\text{Ker}[\dot{V}(x)] = \{x : x_1 x_2 = 0\}$. In order to prove global attractivity we may apply Lasalle's invariance principle. Any invariant set contained in $\{x : x_1 x_2 = 0\}$ should also belong to $\{x : x_1 x_2 = 0 \text{ and } \dot{x}_1 x_2 + x_1 \dot{x}_2 = 0\}$. Computation of this set, however, yields:

$$\{x : x_1 x_2 = 0 \text{ and } \dot{x}_1 x_2 + x_1 \dot{x}_2 = 0\} = \{x : x_1 x_2 = 0 \text{ and } x_2^2 - x_1 x_2^3 - x_1^2 - x_1^3 x_2 = 0\}$$

$$= \{x : x_1 x_2 = 0 \text{ and } x_2^2 - x_1^2 = 0\} = \{0\}.$$

This proves that the largest invariant set contained in $\text{Ker}[\dot{V}(x)] = \{0\}$ and therefore the origin is globally asymptotically stable.
Typical error: non rigorous Lasalle's argument

c) To compute equilibria of the given system we need to fulfill the following equations:

$$\begin{cases} x_2 + x_1(1 - x_1^2 - 2x_2^2) &= 0 \\ -x_1 &= 0 \end{cases}$$

From the second we see $x_1 = 0$ and, substituting this value into the first, we see $x_2 = 0$. Hence the origin is the only equilibrium. We study its stability by the linearization method. In particular:

$$\left.\frac{\partial f}{\partial x}(x)\right|_0 = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}.$$

The characteristic polynomial of this matrix is $\chi(s) = s^2 - s + 1$, with complex conjugate roots in $\frac{1 \pm \sqrt{3} j}{2}$. These have positive real part, and therefore the local phase portrait is that of an unstable focus.

d) To find an invariant set without equilibria we seek C of the following form:

$$C = \{x : r^2 \leq x_1^2 + x_2^2 \leq R^2\}$$

for suitably chosen values of $0 < r < R$. Taking derivatives of $g(x) = x_1^2 + x_2^2$ along solutions of the system yields:

$$\dot{g}(x) = 2x_1 \dot{x}_1 + 2x_2 \dot{x}_2 = 2x_1^2(1 - x_1^2 - 2x_2^2).$$

Notice that:

$$\dot{g}(x) \leq 2x_1^2(1 - x_1^2 - x_2^2) \leq 0$$

for all x with $x_1^2 + x_2^2 = R^2 \geq 1$. On the other hand:

$$\dot{g}(x) \geq 2x_1^2(1 - 2x_1^2 - 2x_2^2) \geq 0$$

for all x with $x_1^2 + x_2^2 = r^2 \leq 1/2$. Hence, the set C is forward invariant provided we choose $R^2 \geq 1$ and $r^2 \leq 1/2$.
Many students did not carefully show for which values of R and r set C is invariant. Others used inequalities in the wrong way.

e) For initial conditions in C, by virtue of Poincarè-Bendixson's criterion, ω-limit sets are closed curves (corresponding to periodic solutions).

3. Exercise

a) A system is (control) affine, when its equations can be written as $\dot{x} = f(x) + g(x)u$. Therefore, the system is affine with:

$$f(x) = \begin{bmatrix} x_2 + x_3^3 \\ -x_1^3 \\ -x_1^3 \end{bmatrix} \qquad g(x) = \begin{bmatrix} x_2 \\ -x_1^3 \\ x_3 \end{bmatrix}$$

b) Pick $V(x) = \alpha_1 x_1^{n_1} + \alpha_2 x_2^{n_2} + \alpha_3 x_3^{n_3}$. For the system to be lossless and passive we need:

$$\frac{\partial V}{\partial x}(x) \cdot f(x) = 0 \qquad \frac{\partial V}{\partial x}(x) \cdot g(x) = y.$$

The first condition reads:

$$[n_1\alpha_1 x_1^{n_1-1}, n_2\alpha_2 x_2^{n_2-1}, n_3\alpha_3 x_3^{n_3-1}] \cdot f(x) = n_1\alpha_1 x_1^{n_1-1}(x_2 + x_3^3) - n_2\alpha_2 x_2^{n_2-1} x_1^3 - n_3\alpha_3 x_3^{n_3-1} x_1^3 = 0$$

The second condition yields:

$$n_1\alpha_1 x_1^{n_1-1} x_2 - n_2\alpha_2 x_2^{n_2-1} x_1^3 + n_3\alpha_3 x_3^{n_3} = x_3^4.$$

From the latter we see:

$$n_1 - 1 = 3, \qquad n_2 - 1 = 1, \qquad \alpha_1 n_1 = \alpha_2 n_2, \qquad n_3 = 4, \qquad \alpha_3 n_3 = 1.$$

In particular, then, $n_1 = 4$, $n_2 = 2$, $n_3 = 4$, $\alpha_3 = 1/4$ and $4\alpha_1 = 2\alpha_2$. From the first condition, however, $n_1\alpha_1 = n_3\alpha_3$. This implies, $n_1\alpha_1 = 1$, and therefore $\alpha_1 = 1/4$. Finally we see $\alpha_2 = 1/2$. With the above choice of storage function we see that:

$$\frac{\partial V}{\partial x}(x) \cdot [f(x) + g(x)u] = yu.$$

c) We pick a static feedback of the form $u = -y$. This choice leads to:

$$\dot{V}(x) = -y^2 = -x_3^8 \leq 0.$$

It is easy to see that $V(x)$ is positive definite (sum of 3 squares) and radially unbounded (a bound on V implies a bound on each of its terms and therefore on each of the state variables). In order to prove global asymptotic stability we only need to check that the largest invariant set contained in the kernel of $\dot{V}(x)$ is the origin. Notice that $\mathrm{Ker}[\dot{V}(x)] = \{x : x_3 = 0\}$. Any invariant set contained in $\mathrm{Ker}[\dot{V}(x)]$ however is also contained in:

$$\{x : x_3 = 0, \dot{x}_3 = 0\} = \{x : x_3 = 0, -x_1^3 = 0\} = \{x : x_1 = 0, x_3 = 0\}.$$

In turn, any invariant set contained in the latter set is also contained in $\{x : x_3 = 0, x_1 = 0, \dot{x}_1 = 0\} = \{0\}$. Thus $\{0\}$ is the largest invariant set contained in $\mathrm{Ker}[\dot{V}(x)]$ and, by the Lasalle's invariance principle, 0 is a globally asymptotically stable equilibrium.
Typical error: using feedback-linearization and dividing times x_3^4 in order to design the linearizing feedback. This is not allowed as x_3 vanishes at the origin and also in a plane of $\mathbb{R}^3$.

d) For $u = -1$, the second equation becomes:

$$\dot{x}_2 = -x_1^3 + x_1^3 = 0.$$

As a consequence, for any solution $x(t)$ we see that $\dot{x}_2(t) \equiv 0$ for all t such that the solution is defined, and in particular, $x_2(t) = x_2(0)$.

e) Equations for the remaining components read as:

$$\dot{x}_1 = x_3^3 \qquad \dot{x}_3 = -x_1^3 - x_3.$$

We analyze this planar dynamics by considering the Lyapunov function $W(x_1, x_3) = x_1^4/4 + x_3^4/4$. This is radially unbounded and positive definite with respect to x_1 and x_3. Its derivative along solutions read:

$$\dot{W} = x_1^3 x_3^3 + x_3^3(-x_1^3 - x_3) = -x_3^4 \leq 0$$

Hence $\dot{W}$ is negative semi-definite. Moreover, the largest invariant set contained in $\{(x_1, x_3) : x_3 = 0\}$ is also contained in $\{(x_1, x_3) : x_3 = 0, -x_1^3 - x_3 = 0\} = \{0\}$. By Lasalle's invariance principle, any solution in (x_1, x_3) space converges to the origin. Hence, given any $x_0 = [x_{01}, x_{02}, x_{03}] \in \mathbb{R}^3$ we see that $\omega(x_0) = \{[0, x_{02}, 0]'\}$.

4. Exercise

a) To verify the relative degree we keep differentiating the output y until u appears. This yields:

$$\dot{y} = \dot{x}_1 = x_2$$

$$\ddot{y} = \dot{x}_2 = \text{atan}(x_3) + [2 + \sin(x_1 + x_2^2)]u.$$

As we have differentiated y twice, the relative degree is 2, provided the coefficient of u is different from 0 (either locally or globally). Notice that

$$2 + \sin(x_1 + x_2^2) > 0 \qquad \text{for all } x \in \mathbb{R}^3,$$

hence the relative degree is globally defined.

b) The I/O linearizing feedback can be formulated as:

$$u = \frac{v - \text{atan}(x_3)}{2 + \sin(x_1 + x_2^2)}.$$

c) To find the internal dynamics we choose the state $\xi = x_1 + x_3$. With this choice we see that,

$$\dot{\xi} = -(x_1 + x_3)^3 + x_1 - x_2 + x_2 = -\xi^3 + y$$

which is indeed independent of u and only a function of ξ and y, as required. The normal form is the following:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= v \\ \dot{\xi} &= -\xi^3 + y \\ \\ y &= x_1 \end{aligned}$$

d) The internal dynamics $\dot{\xi} = -\xi^3 + y$ is ISS with respect to the input y. In fact, taking $V(\xi) = \xi^2$ we see that:

$$\dot{V} = 2\xi(-\xi^3 + y) = -2\xi^4 + 2\xi y \leq -2|\xi|^4 + 2|\xi||y|$$

Notice that the following implication holds:

$$|\xi| \geq \sqrt[3]{2|y|} \Rightarrow \dot{V} \leq -|\xi|^4$$

which proves V is an ISS-Lyapunov function of type II and therefore the internal dynamics are ISS as claimed.
Typical error: not proving the ISS property in a rigorous way.

e) To stabilize the chain of integrators it is enough to choose $v = -k_1 x_1 - k_2 x_2$ for any positive values k_1 and k_2. This results in the linear stable dynamics: $\ddot{y} = -k_2 \dot{y} - k_1 y$.

f) The feedback achieves global asymptotic stability as the closed-loop system can be seen as a cascade of the y-dynamics, which are asymptotically stable forcing an ISS subsystem (internal dynamics). Hence, global asymptotic stability follows.